

Interim Analyses and Group Sequential Designs

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Introduction

Group-sequential designs schedule a sequence of interim analyses at pre-specified information fractions, each with efficacy and/or futility boundaries. Trials can stop early for benefit, harm, or futility while preserving overall Type I error. They are the standard for confirmatory trials with ethical imperatives for early stopping.

Prerequisites

Type I error; sample-size calculation; information fraction.

Theory

With K analyses and overall two-sided alpha α , boundaries are chosen so the union of rejection events has probability α under the null.

Common families: - **O'Brien-Fleming**: conservative early, near-nominal late – preserves nominal alpha at final analysis. - **Pocock**: constant boundary across looks – more early stopping but harder to reach final. - **Alpha-spending** (Lan-DeMets): flexible timing; spending function $f(t)$ at information fraction t .

Assumptions

Analyses occur at pre-specified information fractions; test statistic is approximately normal at each look.

R Implementation

```
library(rpact)

# Group-sequential design: 3 analyses, O'Brien-Fleming, alpha=0.025 one-sided
design <- getDesignGroupSequential(
  sided = 1, alpha = 0.025, beta = 0.2,
  informationRates = c(0.33, 0.67, 1),
  typeOfDesign = "OF"
)
print(design)

# Corresponding sample sizes for two-mean comparison
ss <- getSampleSizeMeans(design = design,
  alternative = 0.3, stDev = 1)
print(ss)
```

Output & Results

Three-stage design with cumulative alpha budgets per stage summing to 0.025; stage sample sizes scale with the chosen information fractions.

Interpretation

“The group-sequential design with O’Brien-Fleming boundary allocated very little alpha to the first two interims (< 0.001 each), preserving nearly the full alpha for the final analysis. Early stopping is extremely unlikely unless the effect is large.”

Practical Tips

- OF boundaries are standard for confirmatory trials; Pocock may suit exploratory ones or single-arm phase II.
- Alpha-spending (Lan-DeMets) is more flexible – timing does not need to be exact, only pre-specified.
- Independent DMC must oversee interim analyses; investigators remain blinded.
- Adjust point estimates and CIs for stopping (median-unbiased, repeated CI).
- Futility boundaries (betagamma spending) complement efficacy and can be non-binding.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale